3D Matrices: Factorisation of Determinants

Questions 1–31

Question 1. It is given that $\mathbf{A} = \begin{pmatrix} 4 & 2 & 6 \\ 0 & 6 & 15 \\ 0 & 0 & 12 \end{pmatrix}$.

a) Find $|\mathbf{A}|$.

b) Find $2\begin{vmatrix} 2 & 1 & 3 \\ 0 & 6 & 15 \\ 0 & 0 & 12 \end{vmatrix}$.

c) Find $6\begin{vmatrix} 2 & 1 & 1 \\ 0 & 6 & 5 \\ 0 & 0 & 4 \end{vmatrix}$.

d) Find $12\begin{vmatrix} 1 & 1 & 1 \\ 0 & 6 & 5 \\ 0 & 0 & 4 \end{vmatrix}$.

e) Find $48\begin{vmatrix} 1 & 1 & 1 \\ 0 & 6 & 5 \\ 0 & 0 & 1 \end{vmatrix}$.

Question 2. It is given that $\mathbf{A} = \begin{pmatrix} 22 & 0 & 0 \\ 660 & 15 & 0 \\ 220 & -12 & 2 \end{pmatrix}$.

a) Find $|\mathbf{A}|$.

b) Find $2\begin{vmatrix} 22 & 0 & 0 \\ 660 & 15 & 0 \\ 220 & -12 & 1 \end{vmatrix}$.

c) Find $4\begin{vmatrix} 11 & 0 & 0 \\ 330 & 15 & 0 \\ 110 & -12 & 1 \end{vmatrix}$.

d) Find $12\begin{vmatrix} 11 & 0 & 0 \\ 110 & 5 & 0 \\ 110 & -12 & 1 \end{vmatrix}$.

e) Find $60\begin{vmatrix} 11 & 0 & 0 \\ 22 & 1 & 0 \\ 110 & -12 & 1 \end{vmatrix}$.

f) Find $660\begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 10 & -12 & 1 \end{vmatrix}$.

g) Hence write 660 as a product of its prime factors.

Question 3. It is given that $\mathbf{A} = \begin{pmatrix} k & 3k^3 & 4k^3(k-1) \\ 0 & k^2 & 2k^2(k-1) \\ 0 & 0 & 3k^2(k-1) \end{pmatrix}$ where k is a constant.

a) Find $|\mathbf{A}|$.

b) Find $k\begin{vmatrix} 1 & 3k^2 & 4k^2(k-1) \\ 0 & k^2 & 2k^2(k-1) \\ 0 & 0 & 3k^2(k-1) \end{vmatrix}$.

c) Find $k(k-1)\begin{vmatrix} 1 & 3k^2 & 4k^2 \\ 0 & k^2 & 2k^2 \\ 0 & 0 & 3k^2 \end{vmatrix}$.

d) Find $k^3(k-1)\begin{vmatrix} 1 & 3k^2 & 4 \\ 0 & k^2 & 2 \\ 0 & 0 & 3 \end{vmatrix}$.

e) Find $k^5(k-1)\begin{vmatrix} 1 & 3 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{vmatrix}$.

f) Hence write $|\mathbf{A}|$ in fully factorised form.

Question 4. For constant k, it is given that

$$\mathbf{A} = \begin{pmatrix} 5k-10 & 0 & 0 \\ 4k-8 & 6-2k & 0 \\ 12-6k & 5k-15 & 3k-18 \end{pmatrix}.$$

a) Find $|\mathbf{A}|$.

b) Find $(k-2)\begin{vmatrix} 5 & 0 & 0 \\ 4 & 6-2k & 0 \\ -6 & 5k-15 & 3k-18 \end{vmatrix}$.

c) Find $(k-2)(k-3)\begin{vmatrix} 5 & 0 & 0 \\ 4 & -2 & 0 \\ -6 & 5 & 3k-18 \end{vmatrix}$.

d) Find $(k-2)(k-3)(k-6)\begin{vmatrix} 5 & 0 & 0 \\ 4 & -2 & 0 \\ -6 & 5 & 3 \end{vmatrix}$.

e) Hence write $|\mathbf{A}|$ in fully factorised form.

Question 5. It is given that $\mathbf{A} = \begin{pmatrix} 24k & 4k+4 & 3k-6 \\ -6k & 2k+2 & 2k-4 \\ 12k & -3k-3 & 8-4k \end{pmatrix}$ where k is a constant. Completely factorise $|\mathbf{A}|$ without evaluating a determinant unless it is completely independent of k.

Question 6. It is given that

$$\mathbf{A} = \begin{pmatrix} 4k-2 & 8k-4 & 5-10k \\ 3-9k & 18k-6 & 6k-2 \\ -1-4k & 12k+3 & 20k+5 \end{pmatrix}$$

where k is a constant. By factorising $|\mathbf{A}|$, solve the equation $|\mathbf{A}| = 0$.

Question 7. It is given that $\mathbf{A} = \begin{pmatrix} 0 & -1 & 3 \\ 2 & 2 & 3 \\ 2 & 1 & -3 \end{pmatrix}$.

a) Find $|\mathbf{A}|$.

b) Let R_1, R_2 and R_3 be the first, second and third rows of $|\mathbf{A}|$ respectively. Perform the following steps in order:

- Let the new $R_1 = R_1 + R_3$.
- Let the new $R_2 = R_2 + R_3$.

c) Use your answer to b) to find $|\mathbf{A}|$.

Question 8. It is given that $\mathbf{A} = \begin{pmatrix} 2 & 2 & -1 \\ 2 & 5 & 0 \\ 2 & 5 & 4 \end{pmatrix}$.

a) Find $|\mathbf{A}|$.

b) Let R_1, R_2 and R_3 be the first, second and third rows of $|\mathbf{A}|$ respectively, and let C_1 be the first column of $|\mathbf{A}|$. Perform the following steps in order:

- Factor 2 out of C_1.
- Let the new $R_3 = R_3 - R_2$.
- Let the new $R_2 = R_2 - R_1$.

c) Use your answer to b) to find $|\mathbf{A}|$.

Question 9. By factoring out constants, and by using row and column operations, reduce the determinants below to the following format:

$$\begin{vmatrix} \# & \# & \# \\ 0 & \# & \# \\ 0 & 0 & \# \end{vmatrix}$$

and hence calculate the determinant.

a) $\begin{vmatrix} 1 & 0 & 5 \\ 2 & 1 & 2 \\ 3 & 2 & 1 \end{vmatrix}$

b) $\begin{vmatrix} 2 & 4 & 6 \\ 3 & -1 & 1 \\ -2 & 2 & 1 \end{vmatrix}$

c) $\begin{vmatrix} 4 & 1 & 6 \\ -1 & 3 & 3 \\ 4 & -2 & 3 \end{vmatrix}$

d) $\begin{vmatrix} 3 & 2 & 5 \\ -2 & 4 & 2 \\ -4 & -1 & 9 \end{vmatrix}$

Question 10. It is given that $\mathbf{A} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -2 & 3 \\ 0 & 0 & 5 \end{pmatrix}$.

a) Find $|\mathbf{A}|$. Consider rotating the rows:

b) Find $\begin{vmatrix} 0 & 0 & 5 \\ 2 & 1 & 3 \\ 0 & -2 & 3 \end{vmatrix}$.

c) Find $\begin{vmatrix} 0 & -2 & 3 \\ 0 & 0 & 5 \\ 2 & 1 & 3 \end{vmatrix}$. Consider rotating the columns:

d) Find $\begin{vmatrix} 1 & 3 & 2 \\ -2 & 3 & 0 \\ 0 & 5 & 0 \end{vmatrix}$.

e) Find $\begin{vmatrix} 3 & 2 & 1 \\ 3 & 0 & -2 \\ 5 & 0 & 0 \end{vmatrix}$. Consider swapping two rows or two columns:

f) Find $\begin{vmatrix} 0 & -2 & 3 \\ 2 & 1 & 3 \\ 0 & 0 & 5 \end{vmatrix}$.

g) Find $\begin{vmatrix} 1 & 2 & 3 \\ -2 & 0 & 3 \\ 0 & 0 & 5 \end{vmatrix}$.

h) What do you notice?

Question 11. For constants $a, b, c, d, e, f, g, h, i$, it is given that $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = 6$. Write down the value of $\begin{vmatrix} g & h & i \\ d & e & f \\ a & b & c \end{vmatrix}$.

Question 12. By reorganising the rows and/or columns only, rewrite the determinants below in the following format:

$$\begin{vmatrix} \# & \# & \# \\ 0 & \# & \# \\ 0 & 0 & \# \end{vmatrix}$$

and hence calculate the determinant.

a) $\begin{vmatrix} 3 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 5 & 0 \end{vmatrix}$

b) $\begin{vmatrix} 0 & 0 & 2 \\ 0 & 4 & 0 \\ -5 & 0 & 0 \end{vmatrix}$

c) $\begin{vmatrix} 0 & 0 & 6k \\ k & 0 & 0 \\ 0 & 4k & 0 \end{vmatrix}$, where k is a constant.

d) $\begin{vmatrix} 0 & pq & 0 \\ p+q & 0 & 0 \\ 0 & 0 & pq \end{vmatrix}$, where p and q are constants.

Question 13. Show clearly, without calculation, that each of the following determinants are equal to zero:

a) $\begin{vmatrix} 1 & 1 & 1 \\ 3 & 4 & -5 \\ 1 & 1 & 1 \end{vmatrix}$

b) $\begin{vmatrix} 2 & -3 & 4 \\ 3 & 0 & 6 \\ 4 & 2 & 8 \end{vmatrix}$

Question 14. It is given that $\mathbf{A} = \begin{pmatrix} 4k+1 & k & 4k+1 \\ 3k & k & 3k \\ -2 & 4 & 2k+2 \end{pmatrix}$ where k is a constant. Let R_1 and R_2 be the first and second rows of $|\mathbf{A}|$ respectively, and let C_1 and C_3 be the first and third columns of $|\mathbf{A}|$. Perform the following steps in order:

a) Let the new $R_1 = R_1 - R_2$; let the new $C_3 = C_3 - C_1$.

b) Use your answer to a) to find a fully factorised form for $|\mathbf{A}|$.

Question 15. Show that each of these are independent of the constant p:

a) $\begin{vmatrix} 1 & p & -p \\ 1 & p+2 & 1 \\ -1 & -p & p+1 \end{vmatrix}$

b) $\begin{vmatrix} p-1 & 1 & p-1 \\ p-2 & 1 & p-2 \\ 3p-3 & p & 3p \end{vmatrix}$

c) $\begin{vmatrix} 2 & -p & -1 \\ 1 & 2p-5 & 2 \\ p & 2 & 1 \end{vmatrix}$

d) $\begin{vmatrix} p-2 & 3p-3 & p-1 \\ p+3 & p+14 & p+3 \\ p-1 & 3p-2 & p \end{vmatrix}$

Question 16. Fully factorise each of these determinants, where n is a constant:

a) $\begin{vmatrix} n+5 & 2n+4 & 9-n \\ 4-n & 4-n & 8-3n \\ n+3 & 2n+2 & 5 \end{vmatrix}$

b) $\begin{vmatrix} -2n & 1-3n & 1 \\ -4-n & 5-2n & 5 \\ -1-3n & 2-2n & n+2 \end{vmatrix}$

c) $\begin{vmatrix} 2n-4 & 3n-6 & 4-2n \\ -6 & 4n-6 & 2-5n \\ 3n-6 & 3n-6 & 2-n \end{vmatrix}$

d) $\begin{vmatrix} 9-15n & 12-17n & 12-15n \\ 40-49n & 40-49n & 40-50n \\ 13-20n & 16-22n & 16-20n \end{vmatrix}$

Question 17. Riley is attempting to create his own question for a friend to revise from. He starts with $n^2(n+1)\begin{vmatrix}1 & 2 & -3\\ 0 & 1 & 0\\ 0 & 3 & 1\end{vmatrix}$. His working and annotations are shown below:

1. Multiply C_1 by $(n+1)$, and multiply R_2 by n, and multiply R_3 by n:
$$= \begin{vmatrix}n+1 & 2 & -3\\ 0 & n & 0\\ 0 & 3n & n\end{vmatrix}.$$
2. Let the new $C_3 = C_3 + C_1$:
$$= \begin{vmatrix}n+1 & 2 & -3\\ 0 & n & 0\\ n+1 & 3n+2 & n-3\end{vmatrix}.$$
3. Let the new $C_3 = C_3 + C_2$:
$$= \begin{vmatrix}n+1 & 2 & -1\\ 0 & n & n\\ n+1 & 3n+2 & 4n+1\end{vmatrix}.$$
4. Let the new $C_1 = C_1 - C_2$:
$$= \begin{vmatrix}n-1 & 2 & -1\\ -n & n & n\\ -2n-1 & 3n+2 & 4n+1\end{vmatrix}.$$

a) Identify where Riley has made a mistake.
b) Factorise the determinant in stage (4) of Riley's working as far as possible.

Question 18. Fully factorise each of these determinants, where n is a constant:

a) $\begin{vmatrix}n+1 & n^2 & 1\\ -n-1 & n+1-n^2 & n\\ n+1 & 2n^2-n & 2n^2-n\end{vmatrix}$

b) $\begin{vmatrix}3n-2 & n^2+2n & 2n-1\\ 3n^2-1 & n+2 & 3n^2-1\\ n^2+4n-4 & n^2+2n & n^2+3n-3\end{vmatrix}$

c) $\begin{vmatrix}3n-n^2 & n+1 & n^2-n-6\\ 6+n-n^2 & -n^2-n-2 & n^2-9\\ 6-2n & -n^2-3n-2 & n-3\end{vmatrix}$

d) $\begin{vmatrix}n-7 & n-12 & n-10\\ 5n^2-20n & 11n^2-44n & 10n^2-40n\\ 3n-6 & 5n-10 & 5n-8\end{vmatrix}$

Question 19. Jessica is attempting to factorise the determinant of the matrix

$$\mathbf{A} = \begin{pmatrix} 5p+q & -2q & 3p-3q \\ 9p & 9q & 6p-6q \\ 3pq+3q^2 & -3q^2 & pq-q^2 \end{pmatrix}$$

where p and q are constants. The beginnings of her working and annotations are shown below:

$$\begin{vmatrix} 5p+q & -2q & 3p-3q \\ 9p & 9q & 6p-6q \\ 3pq+3q^2 & -3q^2 & pq-q^2 \end{vmatrix}$$

(1) Factor q out of C_2:

$$= q \begin{vmatrix} 5p+q & -2 & 3p-3q \\ 9p & 9 & 6p-6q \\ 3pq+3q^2 & -3q & pq-q^2 \end{vmatrix}$$

(2) Let the new $C_1 = C_1 - C_3$:

$$= q \begin{vmatrix} 2p+4q & -2 & 3p-3q \\ 3p+6q & 9 & 6p-6q \\ 2pq+4q^2 & -3q & pq-q^2 \end{vmatrix}.$$

Complete Jessica's working to write $|\mathbf{A}|$ in fully factorised form.

Question 20. Fully factorise each of these determinants, where p, q and r are constants:

a) $\begin{vmatrix} 1 & p & pqr \\ 2 & q & pqr \\ 3 & r & pqr \end{vmatrix}$

b) $\begin{vmatrix} 1 & p & q+r \\ 2 & q & p+r \\ 3 & r & p+q \end{vmatrix}$

c) $\begin{vmatrix} 1 & p & pr \\ 2 & q & qr \\ 3 & r & pq \end{vmatrix}$

Question 21. Fully factorise each of these determinants, where p, q and r are constants:

a) $\begin{vmatrix} p-q-r & p-q-r & p-q-r \\ p-q-r & 2p-q-r & p+3q-r \\ p-q-r & p-q & p-q \end{vmatrix}$

b) $\begin{vmatrix} -5q-7r & -3p-3q+3r & 2p-2q-4r \\ 3p-q-3r & p+q-r & 2p-2q-4r \\ p-6q-9r & -3p-3q+3r & 3p-3q-6r \end{vmatrix}$

c) $\begin{vmatrix} 2p+q+r & p-q & p+r \\ -2p+2r & -p+q-r & -2p-2r \\ 3p+q+2r & 2p+r & 2p+2r \end{vmatrix}$

d) $\begin{vmatrix} 2p-4q+r & 3p-3q & 3p-6q+3r \\ p-q & p-q & p-3q+2r \\ p-3q+r & 2p-2q & 2p-2q \end{vmatrix}$

Question 22. Fully factorise each of these determinants, where p, q and r are constants:

a) $\begin{vmatrix} p & p^2 & p^3 \\ q & q^2 & q^3 \\ r & r^2 & r^3 \end{vmatrix}$

b) $\begin{vmatrix} 1 & p & p^3 \\ 1 & q & q^3 \\ 1 & r & r^3 \end{vmatrix}$

c) $\begin{vmatrix} 1 & p^2 & p^3 \\ 1 & q^2 & q^3 \\ 1 & r^2 & r^3 \end{vmatrix}$

d) $\begin{vmatrix} 1 & p & p^2(p+1) \\ 1 & q & q^2(q+1) \\ 1 & r & r^2(r+1) \end{vmatrix}$

Question 23. By first showing that $(p+q+r)$ is a factor where p, q and r are constants, fully factorise each of these:

a) $\begin{vmatrix} 1 & p & q+r \\ 2 & q & p+r \\ 3 & r & p+q \end{vmatrix}$

b) $\begin{vmatrix} 1 & p+q & r \\ 2 & q+r & p \\ 4 & p+r & q \end{vmatrix}$

c) $\begin{vmatrix} 1 & 3 & 6 \\ p+2r & q+2p & r+2q \\ q-r & r-p & p-q \end{vmatrix}$

d) $\begin{vmatrix} p-2r & q-2p & r-2q \\ 2 & 4 & 8 \\ q+3r & r+3p & p+3q \end{vmatrix}$

Question 24. Given that $|\mathbf{A}| = xy \begin{vmatrix} 1 & 2 & 5 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{vmatrix}$, show clearly, by using row and column operations, that

$$|\mathbf{A}| = \begin{vmatrix} 1+5y & 2x & 5y \\ -3y & x & -3y \\ 4y & -x & 4y \end{vmatrix}.$$

Question 25. Given that $|\mathbf{A}| = xy(x-1)(y-1)\begin{vmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{vmatrix}$, show clearly, by using row and column operations, that

$$|\mathbf{A}| = \begin{vmatrix} xy-x & y & xy-x+y \\ x-xy & x-y-1 & x-xy-y \\ x-xy & 0 & x-xy \end{vmatrix}.$$

Question 26. It is given that $\mathbf{A}$ is a singular matrix such that

$$\det \mathbf{A} = \begin{vmatrix} 6 & 3x+3a+5 & -5 \\ 5 & 4x+4a+4 & -4 \\ 4a-x+1 & 3a-2x+1 & x-4a-1 \end{vmatrix}$$

where a is a constant. Find the possible values of x in terms of a.

Question 27. It is given that $\mathbf{A}$ is a singular matrix such that

$$\det \mathbf{A} = \begin{vmatrix} -2a & 6x+24a & -x-a \\ 2x-4a & 7x+28a & 3x-5a \\ 3x-7a & 13x+52a & 4x-8a \end{vmatrix}$$

where a is a constant. Find the possible values of x in terms of a.

Question 28. It is given that $\mathbf{A}$ is a singular matrix such that $\det \mathbf{A} = \begin{vmatrix} 0 & 1 & x-p \\ x & p & px \\ p & q & p^2 \end{vmatrix}$ where p and q are constants.

a) Explain why $\det \mathbf{A} = \begin{vmatrix} 0 & 1 & x-p \\ x & p & px \\ p & q & 0 \end{vmatrix}$.

b) Hence express the possible values of x in terms of p and q.

Question 29. Given p and q are constants, fully factorise

$$\begin{vmatrix} x^3 & 0 & p^2x \\ q & pq & q \\ x+q & p^2+pq & p+q \end{vmatrix}.$$

Question 30. Given p and q are constants, fully factorise

$$\begin{vmatrix} q-x & -p^2 & x-q \\ x-q & px & q-p \\ x-q & p^2 & p+2q-x \end{vmatrix}.$$

Question 31.

a) Given a and b are constants, fully factorise

$$\begin{vmatrix} a-2x+b & x+a & 4a-2x+b \\ a-2x+2b-1 & x+a+1 & 4a-2x+3b \\ 3a+b-x-1 & 1 & 6a+2b-2x \end{vmatrix}.$$

b) Transformation T can be represented by

$$\mathbf{A} = \begin{pmatrix} 13-2x & x+6 & 31-2x \\ 19-2x & x+7 & 45-2x \\ 24-x & 1 & 50-2x \end{pmatrix}.$$

When a 3D shape of volume $300\,\text{cm}^3$ is transformed by T, its new volume is $1260\,\text{cm}^3$. Use your answer to part a) to find the possible values of x.